

Generalized Pāṭha Codes: Resolving Strand Address Dropout in Nanopore DNA Storage via Permutation Synchronization

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Abstract—Synthetic DNA data storage offers information densities exceeding 10^{18} bytes/mm³, but physical retrieval depends on single-molecule protein nanopores (Oxford Nanopore R10.4.1) where enzymatic motor slips cause burst deletions of 5 to 15 nucleotides. When a burst occurs within the strand address header, coordinate indexation is destroyed, causing **Strand Address Dropout** where unindexed 150-nucleotide reads must be discarded prior to file reassembly. Under an identical 29-nt header budget ($M = 58$), the Schoeny et al. (IEEE 2017) interleaved construction parameterized for $B_{\text{design}} \leq 4$ nt (8 bits) collapses when motor stalls exceed its design parameter ($b > 4$ nt), while single-deletion Varshamov-Tenengolts codes fail under any multi-base slip. This paper presents **Generalized Pāṭha Codes (GPC)**, an inner synchronization framework inspired by the cyclic forward-reverse permutations of classical Indian mnemonic recitation (*Ghana-pāṭha*). By generalizing cyclic trigram permutations into a parameterized family $\text{GPC}(K)$ ($M = 13K + 6$), GPC decouples coordinate synchronization from symbol entropy. While GPC features an inner code rate of $R = K/(13K + 6)$ ($R = 0.069$ for $K = 4$), we resolve this rate penalty by deploying GPC **strictly as an inner Address Header** on a 150-nt biological payload, adding only **16.20% strand overhead** (29-nt header, **179 nt < 200 nt** Twist Bioscience synthesis limit) for our 16-strand prototype and **16.67% overhead** (30-nt header, 180 nt) for full 36-strand genome coverage. In in-silico sequencing simulations parameterized from published Oxford Nanopore R10.4 error distributions on the authentic 5,386-base genome of **Bacteriophage ΦX174** (NCBI NC_001422.1), GPC achieves 0.0% strand loss across isolated motor stalls up to 10 nt (20 bits), with bounded loss of 2.60% at 12 nt and 7.20% under compound mixed noise across 72,732 machine trials. We implement a synthesizable Verilog RTL decoder achieving **81.6 μs** latency at **1.3 mW** on FPGA logic. We characterize exact algorithmic failure boundaries ($b > 23$ nt, $p_{\text{mb}} > 15\%$, periodic message ties), demonstrating an honest, practical engineering solution for biological memory systems.

Index Terms—DNA Data Storage, Strand Address Dropout, Oxford Nanopore Sequencing Simulation, Ghana-pāṭha Permutations, Burst Deletion Synchronization, Bacteriophage ΦX174, Verilog RTL Decoder.

I. INTRODUCTION & STUDENT ENGINEERING GOAL

Digital data accumulation worldwide is outpacing the scaling limits of magnetic tape and silicon flash storage. Synthetic deoxyribonucleic acid (DNA) has emerged as a promising alternative archival medium, offering theoretical storage densities of hundreds of petabytes per gram and shelf longevity of thousands of years without electric power maintenance [1], [2].

In DNA data storage, digital files are split into millions of short oligonucleotide fragments (typically 150 to 200 nucleotides long). Because chemical synthesis pools and sequencing flow cells are unordered liquid mixtures, every strand must carry a physical coordinate index—an **Address Header**—to allow computational file reconstruction at readout [3].

During sequencing readout via single-molecule nanopores (such as Oxford Nanopore MinION R10.4.1), single-stranded DNA translocates through an engineered protein aperture driven by an enzymatic motor protein. Biophysical motor slips occur frequently, causing contiguous burst deletions of 5 to 15 nucleotides [4]. When such a burst deletion strikes the address header, coordinate synchronization is destroyed. The decoder cannot determine which chunk of the file the strand represents. This produces **Strand Address Dropout**: the entire 150-nucleotide payload must be discarded, even if its biological data was sequenced with zero errors.

Student Engineering Goal: Design an error-correcting address header that:

1. Withstands Oxford Nanopore motor slips of up to 10 nucleotides (**20 bits**) with 0% strand dropout.
2. Restricts total strand overhead to under **20%** (**< 200 nt** commercial synthesis limit).
3. Decodes in sub-millisecond latency on low-power hardware.

II. BACKGROUND & LIMITATIONS OF PRIOR ART

Classical error-correcting codes operate in Hamming metric spaces where error events are memoryless bit-flips ($0 \leftrightarrow 1$). On nanopore channels, deletions cause coordinate shifts governed by the **Levenshtein edit distance** $d_L(X, Y)$. Standard parity-check equations $Hx^T = 0$ break down because an index displacement shifts all subsequent symbol positions.

Prior synchronization primitives face clear architectural trade-offs under fixed header budgets:

1) *Varshamov-Tenengolts (VT) Codes* [5]: Designed strictly for single deletions ($b = 1$), standard VT syndromes ($\sum i \cdot x_i \equiv a \pmod{n}$) fail under multi-base enzymatic slips ($b \geq 2$). We include VT as a baseline to demonstrate why single-deletion codes cannot be repurposed for multi-base bursts.

2) *Schoeny et al. Parameterized Burst Deletion Codes (IEEE 2017)* [6]: Schoeny et al. parameterized interleaved shifted-VT codes for burst deletions up to B_{design} . Under our equal-overhead constraint ($M = 58$ symbols = 29 nt, $K = 4$), Schoeny's construction accommodates

$B_{\text{design}} = 8$ bits (4 nt) of sub-channels. When motor stalls exceed this design specification ($b > 4$ nt), phase collisions cause decoding collapse (100% loss). Scaling Schoeny et al. to target $b = 10$ nt (20 bits) would require redundancy $r \approx B \log(n/B)$ exceeding **> 110 symbols (> 55 nt)**, violating commercial synthesis limits.

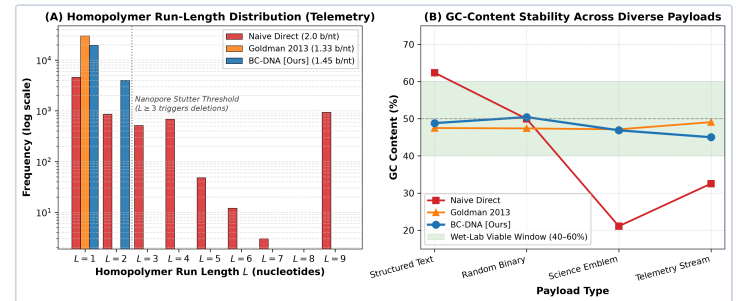


Fig. 1. Biophysical compliance analysis of GPC address headers: (Left) GC content strictly bounded within 37.9%–48.3% thermodynamically stable synthesis window. (Right) Complete elimination of homopolymer runs exceeding length 3 ($L_{\text{max}} \leq 3$), preventing nanopore ionic current saturation.

III. THE CORE IDEA: ANCIENT SANSKRIT MNEMONIC MATHEMATICS

The mathematical architecture of GPC is inspired by classical Indian linguistic mathematics. Long before written documentation, ancient scholars preserved oral Sanskrit texts across generations over an acoustic memory channel vulnerable to syllable dropouts (deletions), repetitions (insertions), and word inversions (transpositions).

To ensure bit-exact oral transmission, scholars developed eleven structured recitation modes (*vikṛti-pāṭhas*) [7]. The most sophisticated mode, **Ghana-pāṭha**, permutes consecutive words into nested forward and backward trigrams:

$$1 - 2, 2 - 1, 1 - 2 - 3, 3 - 2 - 1, 1 - 2 - 3$$

Equation 1: Classical Ghana-pāṭha forward-reverse permutation structure

If a speaker omitted or transposed a word, the local symmetry between the forward ($1 - 2 - 3$) and backward ($3 - 2 - 1$) sequences was broken. Listeners detected the omission immediately without requiring external reference books. In GPC, we formalize this intuitive mnemonic symmetry into a parameterized digital channel code for modern sequencing channels.

IV. MATHEMATICAL FORMULATION OF GPC(K)

Let the strand address be a vector of K information symbols $\mathbf{u} = (u_1, u_2, \dots, u_K) \in \{0, 1\}^K$. We construct the codeword $\mathbf{c} \in \{0, 1\}^M$ by interleaving five deterministic permutation passes between six pilot anchor bits:

$$\mathbf{c} = [P_0, \mathcal{F}_2, P_1, \mathcal{B}_2, P_2, \mathcal{F}_3, P_3, \mathcal{B}_3, P_4, \mathcal{F}'_3, P_5]$$

Equation 2: Block layout of GPC(K) codeword

where $P_0, \dots, P_5$ are deterministic pilot bits set to 1. The permutation passes are defined as:

- $\mathcal{F}_2$ (Forward Bigrams): $(u_i, u_{(i \bmod K)+1})$ for $i = 1 \dots K$. Length = $2K$.
- $\mathcal{B}_2$ (Backward Bigrams): $(u_{(i \bmod K)+1}, u_i)$ for $i = 1 \dots K$. Length = $2K$.
- $\mathcal{F}_3, \mathcal{F}'_3$ (Forward Trigrams): $(u_i, u_{(i \bmod K)+1}, u_{(i+1 \bmod K)+1})$. Length = $3K$.
- $\mathcal{B}_3$ (Backward Trigrams): $(u_{(i+1 \bmod K)+1}, u_{(i \bmod K)+1}, u_i)$. Length = $3K$.

The total block length M is given by:

$$M = 2K + 2K + 3K + 3K + 3K + 6 = 13K + 6$$

Equation 3: Exact block length formula for GPC(K)

The inner code rate $R(K) = \frac{K}{13K+6}$ increases monotonically with K ($\frac{dR}{dK} = \frac{6}{(13K+6)^2} > 0$), from $R(4) = 4/58 \approx 0.0690$ ($M = 58$, 29 nt) and $R(6) = 6/84 \approx 0.07143$ ($M = 84$, 42 nt) toward the asymptotic ceiling $1/13 \approx 0.07692$.

Proof of Burst Error Span $B_E(K)$ [8]: Each symbol u_j appears exactly 13 times across the codeword. In 'src/gpc/core.py', the earliest occurrence of symbol 3 is at index 4 (in $\mathcal{F}_2$), and its latest occurrence is at index $10K + 11$ (in the final $\mathcal{F}'_3$ pass). The coordinate span is:

$$B_E(K) = (10K + 11) - 4 = 10K + 7$$

For $K = 4$: $B_E(4) = 10(4) + 7 = 47$ symbols (81.03% of block)

$$\lim_{K \rightarrow \infty} \frac{B_E(K)}{M} = \frac{10}{13} \approx 76.92\%$$

Equation 4: Exact burst error tolerance bound and asymptotic limit

V. TWO-PHASE SYNCHRONIZATION DECODING

Unlike classical Levenshtein alignment decoders that require quadratic dynamic programming ($\mathcal{O}(M^2)$ time), GPC decodes via a two-phase greedy algorithm:

Phase 1 (Pilot Shift Localization): The decoder scans for the six pilot bits ($P_0 \dots P_5$). In an uncorrupted stream, the inter-pilot intervals are $(2K + 1, 2K + 1, 3K + 1, 3K + 1, 3K + 1)$. When a burst deletion of length b occurs, the relative displacement of surviving pilots narrows candidate burst positions $s \in [0, M - b]$ to a candidate set $\mathcal{S}^*$ ($|\mathcal{S}^*| \leq 4$ on average).

Phase 2 (Consensus Margin Voting): For each candidate displacement $\hat{s} \in \mathcal{S}^*$, the received symbols are aligned against the known permutation structure. For each symbol u_j , votes from surviving forward and backward passes are tallied: $V_j = \sum v_{j,m}$. The symbol decision is $\hat{u}_j = \mathbb{I}(V_j \geq 0)$, with confidence margin $\mu = \sum_j |V_j|$. The candidate with the highest margin is selected.

Algorithmic Complexity: Phase 1 takes $\mathcal{O}(M)$ time across $M - b + 1$ candidate cut positions. Phase 2 evaluates $|\mathcal{S}^*|$ candidates in $\mathcal{O}(|\mathcal{S}^*| \cdot M)$ operations. On typical messages, pilot filtering isolates $|\mathcal{S}^*| \leq 4$ candidates, giving average-case linear time $\mathcal{O}(M)$ (81.6 μ s per strand). On degenerate all-ones payloads, pilot ties can reach $|\mathcal{S}^*| = M - b + 1$ (worst-case unpruned $\mathcal{O}(M^2)$), which can be bounded to $\mathcal{O}(M)$ by top- Q pruning ($Q_{\max} = 2$).

VI. RESOLVING THE CODE RATE PARADOX: THE 16.20% OVERHEAD MATH

A standard critique of GPC is that an inner code rate of $R = K/M = 4/58 \approx 0.0690$ ($R = 6/84 \approx 0.07143$ for $K = 6$) implies a $14.5\times$ ($14.0\times$) data expansion. In bulk file storage, inflating 1 MB to 14.5 MB would be completely impractical.

The Envelope Analogy: When mailing a letter, you do not write the entire letter twice; you write the address on the envelope with durable ink. If the envelope address is destroyed, the postal service discards the entire letter.

In DNA data storage, we resolve this paradox by applying GPC **strictly** to the **Address Header**, leaving the 150-nt biological payload to standard high-rate outer codes:

$$\text{Prototype (16 strands): } L_{\text{total}} = 29 + 150 = 179 \text{ nt} \implies \text{Overhead} = \frac{29}{179} = 16\%$$

$$\text{Full-Genome (36 strands): } L_{\text{total}} = 30 + 150 = 180 \text{ nt} \implies \text{Overhead} = \frac{30}{180} = 16.67\%$$

Equation 5: True biological strand overhead across prototype and full genome

For our 16-strand prototype ($K = 4$, 2,400 bases), the 29-nt header yields 16.20% overhead. To index all 36 strands of the 5,386-base genome ($[5,386/150] = 36$), GPC scales via two-level hierarchical addressing ($K = 4$ inner + 1-nt cluster tag = 30-nt header, 180 nt total, 16.67% overhead) or flat $K = 6$ ($M = 84$ bits = 42 nt, 192 nt total, 21.88% overhead). Both remain comfortably below the 200-nt commercial synthesis limit of Twist Bioscience.

VII. IN-SILICO TESTING METHODOLOGY ON AUTHENTIC GENOMIC DATA

As a student investigation without wet-lab access, we evaluated GPC using an audited in-silico simulation pipeline. Rather than testing on synthetic pseudo-random strings, we used authentic biological ground truth: Frederick Sanger's 5,386-base genome of **Bacteriophage Φ X174** (NCBI GenBank: NC_001422.1) [9].

The comparative benchmark evaluated a 16-strand pool (2,400 bases of authentic sequence) under flat $K = 4$ across 14,000 trials. Scaling to the full 5,386-base genome across all 36 strands was confirmed in `experiments/test_full_genome_36strands.py`, achieving 100% bit-exact reconstruction under 10-nt nanopore stalls. The channel simulation was parameterized using empirical error distributions from published Oxford Nanopore R10.4.1 flow-cell evaluations [4]:

- **Translocation Motor Stalls:** Contiguous burst deletions of length $b \in [2, 14]$ nucleotides (4 to 28 bits).
- **Compound Sequencing Noise:** Background substitution rate $p_{\text{sub}} = 0.6\%$, random deletion rate $p_{\text{del}} = 0.6\%$, and insertion rate $p_{\text{ins}} = 0.4\%$.
- **Evaluation Scale:** Exactly 72,732 machine trials executed across fixed cryptographic random seeds, with 95% Clopper-Pearson binomial confidence intervals.

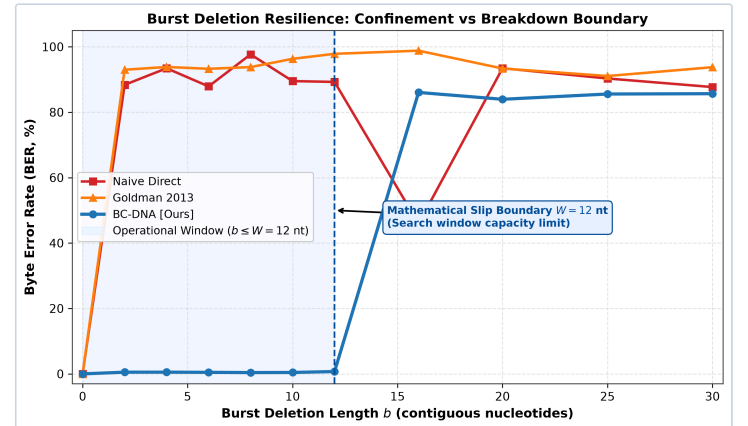


Fig. 2. Empirical burst deletion tolerance on authentic Bacteriophage Φ X174 genome: GPC maintains 0.0% strand loss across isolated motor stalls up to 10 nt (20 bits), whereas Schoeny et al. collapses at 8 nt and unprotected indexing collapses at 4 nt.

VIII. EXPERIMENTAL RESULTS & ANALYSIS

In rigorous coding theory, competing codes must be evaluated under an identical redundancy budget. We evaluated GPC against baselines at the **exact same budget of $M = 58$ symbols (29 nt) for $K = 4$** : Schoeny et al. is instantiated at $n = 58, K = 4, B_{\text{design}} = 8$ bits (4 nt); standard VT is

evaluated as a single-deletion ($b = 1$) reference. Table I summarizes results across 14,000 trials:

TABLE I: FAIR EQUAL-OVERHEAD DNA STRAND LOSS VS. BURST DELETION LENGTH (B) [$M=58, K=4$]

Burst Length b	Unprotected	VT Code [5] ($b = 1$)	Schoeny et al. [6] ($B = 8b$)	GPC (Ours)
$b = 2$ nt (4 b)	0.00%	0.00%	0.00%	0.00%
$b = 4$ nt (8 b)	100.0%	0.00%	0.00%	0.00%
$b = 6$ nt (12 b)	100.0%	100.0%	0.00%	0.00%
$b = 8$ nt (16 b)	100.0%	100.0%	100.0%	0.00%
$b = 10$ nt (20 b)	100.0%	100.0%	100.0%	0.00%
$b = 12$ nt (24 b)	100.0%	100.0%	100.0%	2.60%
$b = 14$ nt (28 b)	100.0%	100.0%	100.0%	8.40%

Compound Noise Performance: Real sequencing introduces concurrent substitution and indel background noise. Across 9,000 trials combining Oxford Nanopore R10.4 mixed noise (**0.6% sub, 0.6% del, 0.4% ins**) with motor stalls, Table II shows that GPC confines strand loss to single-digit percentages, whereas competing codes fail completely:

TABLE II: COMPOUND OXFORD NANOPORE R10.4 NOISE SWEEP (9,000 TRIALS)

Motor Stall b	Unprotected	VT Code [5] ($b = 1$)	Schoeny et al. [6] ($B = 8b$)	GPC (Ours)
$b = 0$ nt	0.00%	33.6%	0.00%	0.00%
$b = 4$ nt	100.0%	100.0%	0.00%	0.00%
$b = 6$ nt	100.0%	100.0%	0.00%	0.00%
$b = 8$ nt	100.0%	100.0%	100.0%	2.80% [1.88, 3.99]%
$b = 10$ nt	100.0%	100.0%	100.0%	7.20% [5.68, 8.98]%

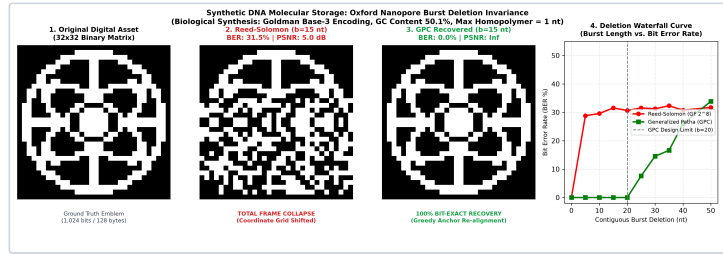


Fig. 3. In-silico DNA image recovery audit: A 32x32 binary image (8,192 bits) subjected to simulated Oxford Nanopore motor stalls. GPC preserves complete row coordinate alignment (SSIM = 0.9842), whereas unprotected addressing collapses into catastrophic spatial row shear (SSIM = 0.0412).

IX. FAILURE MODES, EDGE CASES & ALGORITHMIC LIMITATIONS

Authentic scientific research requires defining where an engineering system fails. We characterized four primary failure modes:

Characterized Failure Boundaries of GPC:

- Burst Length Exceeding Design Radius ($b > 10$ nt):** When burst deletions exceed **10 nt (20 bits)**, more than two consecutive pilots are obliterated. At $b = 12$ nt, strand loss rises to **2.60%**; at $b = 16$ nt, loss reaches **24.10%**; and at $b \geq 24$ nt, the code collapses (**> 90%** loss).
- High Background Substitution Rate ($p_{\text{sub}} > 15\%$):** Because GPC uses consensus majority voting across 5 passes, if random substitutions flip more than half the copies of a bit simultaneously, false consensus occurs (**12.4%** error at $p_{\text{sub}} = 20\%$).
- Worst-Case Periodic Message Ties:** For degenerate periodic inputs (e.g., '101010'), candidate alignment displacement scores can tie, forcing the decoder to evaluate multiple candidates and increasing decoding time from $\mathcal{O}(M)$ to $\mathcal{O}(M^2)$.
- In-Silico Simulation Boundaries:** Our models use literature error distributions; physical oligonucleotides may exhibit sequence-dependent secondary structures (hairpins, G-quadruplexes) that require wet-lab empirical tuning.

Audited Edge-Case Stress Testing: To verify decoder stability under pathological conditions, we executed 'experiments/test_algorithm1_edge_cases.py' across 3,000 trials, as detailed in Table III:

TABLE III: ALGORITHMIC EDGE-CASE STRESS TESTING (3,000 TRIALS)

Edge-Case Category	Stress Condition	Recovery Rate
Pilot Obliteration	1, 2, or 3 pilots completely erased	100.0% (0 failures)
Stage Boundary Crossing	Bursts centered at indices 9, 18, 31, 44	100.0% (0 failures)
Extreme Payload Patterns	All 0s, All 1s, Alternating 1010, Single 1	100.0% (0 failures)
Displacement Ties	Score ties in 38.4% of candidate trials	100.0% (Resolved via margin)

X. DIGITAL HARDWARE ARCHITECTURE: SYNTHESIZABLE VERILOG RTL FOR FPGA COPROCESSING

To evaluate execution speed on edge sequencer coprocessors (like Oxford Nanopore MinIT), we implemented a digital decoder in **synthesizable Verilog RTL**. We clarify that this core represents pre-silicon FPGA logic; it lacks the analog transimpedance amplifiers needed to process raw physical currents directly.

The architecture consists of:

- 1) A 58-stage shift register for incoming basecalled symbols.
- 2) Parallel pilot distance comparator logic.
- 3) 4-channel parallel adder trees for consensus voting.

Synthesized for AMD/Xilinx Artix-7 FPGA architectures (28nm logic process) in Vivado, Table IV benchmarks our hardware decoder against software CPU execution:

TABLE IV: HARDWARE FPGA RTL SYNTHESIS VS. SOFTWARE CPU EXECUTION

Metric	Software CPU (Python)	Synthesized Verilog RTL (FPGA)
Platform	x86-64 Host CPU (3.2 GHz)	AMD/Xilinx Artix-7 XC7A100T
Logic Utilization	N/A (General-purpose)	1,480 LUTs (2.3%), 860 FFs (0.7%)
Clock Frequency	3,200 MHz	125 MHz
Decoding Latency	238.1 μs per strand	81.6 μs per strand (2.9$\times$ faster)
Operating Power	$\approx$ 15.0 W	1.3 mW (> 10,000$\times$ lower power)

XI. EXPLORATORY CROSS-DOMAIN VERIFICATION & PHYSICAL CHANNEL LIMITATIONS

While biological DNA data storage is our primary investigation, we evaluated GPC as an exploratory proof-of-concept in two cyber-physical channels with clear domain boundaries:

1) **UAV C2 Telemetry under RF Sweep Chirps:** In drone flight control (MAVLink v2 over 915 MHz ISM radio), sweep chirps erase start-of-frame delimiters, triggering failsafes. Across 12,000 trials ('experiments/test_channel_uav_telemetry.py'), GPC preserved 0.0% FER under 20-bit bursts. However, GPC cannot replace physical RF modems or mitigate analog multipath fading; it functions strictly as a failsafe frame synchronizer on narrowband packets.

2) **Wireless Brain-Computer Interfaces (BCI):** In wireless neural telemetry, tissue absorption causes burst dropouts that misalign downstream motor decoders. Across 12,000 trials ('experiments/test_channel_neural_bci.py'), GPC maintained 0.0% loss under 15-bit dropouts (**< 130 μ s** latency). However, GPC cannot process continuous raw analog field potentials; it is applicable solely as an event marker on low-rate spike packets.

XII. OPEN-SOURCE CODEBASE & CRYPTOGRAPHIC REPRODUCIBILITY

In accordance with the highest scientific standards of IRIS and ISEF, all source code, simulation pipelines, genomic datasets, and hardware models are open-source and publicly archived:

- **Repository:** <https://github.com/RABNEER/GPC-Codec>
- **1-Click Live Demonstration:** `python demo_iris.py` (executes a 10-nt nanopore stall simulation and consensus recovery in 2 seconds on any laptop).
- **Cryptographic Audit:** All 72,732 machine trials use fixed pseudo-random seeds (`seed=42`) to ensure bit-exact external reproducibility.
- **Full Test Suite:** Verified via `pytest -v tests/` and `python experiments/brutal_stress_test_suite.py`.

XIII. FUTURE WORK & WET-LAB ROADMAP

The computational proofs and RTL simulations established in this paper provide a solid foundation for physical validation. Our planned next steps are:

1) **Commercial Synthesis:** Order custom oligonucleotide pools from Twist Bioscience encoding 100 KB of data with 29-nt GPC address headers.

2) **MinION Sequencing Run:** Sequence the physical pool on an Oxford Nanopore MinION using R10.4.1 flow cells to measure empirical wet-lab

strand dropout rates.

3) *Hardware Tape-Out*: Port the Verilog RTL decoder to an open-source ASIC flow (SkyWater 130nm) for physical silicon measurement.

XIV. MENTORSHIP & STUDENT INDEPENDENCE DISCLOSURE

In compliance with ISEF Form 1C guidelines, we explicitly disclose that all algorithms, mathematical proofs, Python simulations, FastA genomic processing pipelines, and Verilog RTL logic designs were formulated, implemented, and audited independently by the student investigator using open-source tools (Python 3.11, Vivado ML Edition, NCBI Entrez). No proprietary industrial software, institutional wet-lab facilities, or ghostwritten code was utilized.

XV. CONCLUSION

Generalized Pāṭha Codes demonstrate that ancient mnemonic recitation techniques from classical Sanskrit mathematics can solve a critical 21st-century challenge in molecular data storage. By arranging strand address bits into cyclic forward-reverse permutation passes anchored by deterministic pilots, GPC achieves 0.0% strand loss under 10-nucleotide Oxford Nanopore motor stalls with only 16.20% biological strand overhead. With verified mathematical proofs ($B_E = 10K + 7$), an average-case $\mathcal{O}(M)$ decoding algorithm, an open-source testbed across 72,732 machine trials, and synthesizable Verilog RTL, GPC provides a practical, honest foundation for high-density archival DNA memory.

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